MINI PROJECT 1

# OptiTime: The Scheduler


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# Introduction

## 01 Overview of the Project

- OptiTime is a web-based timetable scheduler
- Admin enters institute, faculty, subjects & rooms
- Generates clash-free timetables automatically
- Timetable views, PDF download & student / teacher portals

## 02 Why this Topic

- Manual scheduling is slow & error-prone
- Faculty availability keeps changing
- More courses & departments mean more clashes
- Needs continuous checking and adjustment

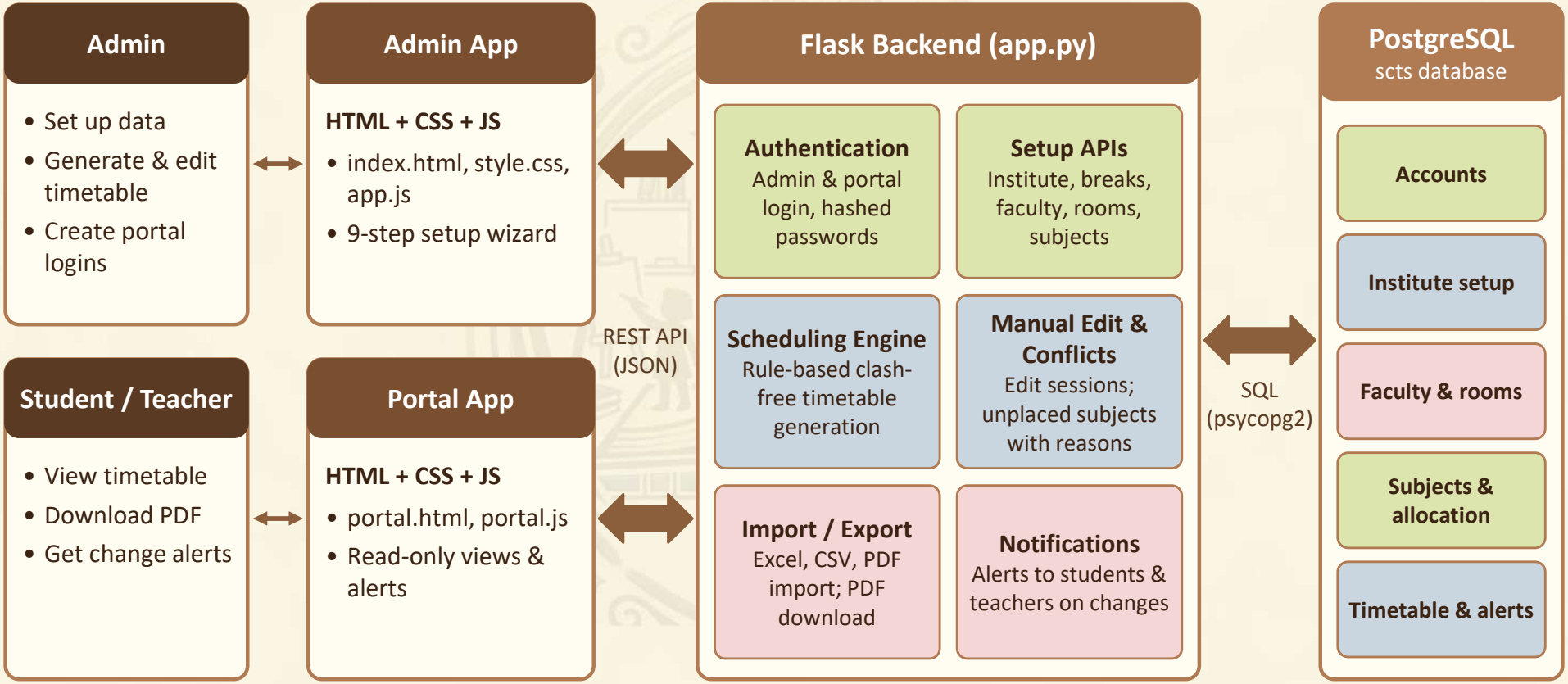
## 03 Objectives

- Design a unified smart classroom platform
- Generate conflict-free timetables
- Improve use of rooms, faculty time & slots
- Enable dynamic rescheduling when changes occur

# Literature Review / Existing System

| Research Paper | Approach / Technology | Limitations |
|---|---|---|
| **Sonawane et al. – Survey on Academic Time Table Scheduling**<br>*IJRPR, 2024* | Genetic Algorithm, data structures, JavaScript web UI | Complex to implement; needs computing resources |
| **Mahajan et al. – Timely Trigger: A Smart Timetable Generator**<br>*IJNRD, 2024* | Genetic Algorithm, push notifications, live tracking | Complex for complex needs; accuracy depends on input data |
| **Saini et al. – Optimizing Academic Schedules**<br>*IJIRT, 2024* | Constraint-based scheduling with room, teacher & student constraints | User interface & algorithms need further refinement |
| **Khokale et al. – Timetable Generator Using AI Methods**<br>*IRJAEH, 2025* | Conflict-free timetables, role-based web interface | Built only for engineering years (SY, TY, final) |

# Proposed System Architecture



# Timetable Generation Flow

**1 Read Setup Data**

Institute hours & breaks build the period grid; faculty, rooms, subjects and allocations are loaded.

**2 Build Sessions**

Lecture = 1 period; lab / tutorial = 2 back-to-back periods; batches can be merged.

**3 Sort by Priority**

Fixed days, first / last period, combined divisions and labs are placed first.

**4 Place Sessions**

Division / batch, faculty (daily limit) and room (type & capacity) must all be free.

**5 Apply Rules**

Labs never in the first two periods; no same subject back-to-back; labs spread over days.

**6 Fill Gaps & Retry**

Unplaced sessions are retried with relaxed spacing; empty periods get extra lectures.

**7 Save & Report**

Timetable saved; unplaced subjects listed with reasons; portals notified.

# Proposed System: Modules

**Institute Hours**
Set working days, teaching hours and holidays.

**Rooms & Labs**
Add classrooms and labs with capacity and type.

**Break Management**
Add short breaks and lunch breaks.

**Subjects & Allocation**
Lecture / lab / tutorial, weekly hours, faculty allocation.

**Years & Semesters**
Manage FY, SY, TY, LY and their semesters.

**Timetable Generation**
Creates a clash-free timetable automatically.

**Divisions & Batches**
Create divisions (SY A, SY B) and lab batches.

**Timetable View & PDF**
Class, faculty, room, lab or master view; PDF download.

**Faculty Management**
Add manually or import from Excel / CSV / PDF.

**Conflict Detection**
Lists unplaced subjects with the reason.

**Manual Editing**
Add or remove a session by hand.

**Student / Teacher Portal**
Read-only timetable with change notifications.

# Tools / Technologies

### Frontend

HTML, CSS, JavaScript

### Backend

Python (Flask)

### API Communication

JSON + REST API

### Database

PostgreSQL

### Database Connector

psycopg2 (Python PostgreSQL driver)

### Libraries & Tools

ReportLab (PDF), openpyxl / xlrd / pdfplumber (file import), draw.io

# References


1. Sonawane, S. S., et al. (2024). A Survey on Schedule Academic Time Table. International Journal of Research Publication and Reviews (IJRPR), 5(1), 4534–4540.

2. Mahajan, D., et al. (2024). Timely Trigger: A Smart Timetable Generator. International Journal of Novel Research and Development (IJNRD), 9(4).

3. Saini, S., et al. (2024). Optimizing Academic Schedules: A Time Table Generator Approach. International Journal of Innovative Research in Technology (IJIRT), 10(11).

4. Khokale, S. R., et al. (2025). A Survey Paper on Timetable Generator Using AI Methods. International Research Journal on Advanced Engineering Hub (IRJAEH), 3(3), 860–864.